

ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC BÁCH KHOA
KHOA KHOA HỌC & KỸ THUẬT MÁY TÍNH



BÁO CÁO BTL Phát triển ứng dụng trên thiết bị di động

Tinner: Nền tảng di động tối ưu hóa quá trình đặt món và địa điểm

Giảng viên: Ths. Hoàng Lê Hải Thanh
Lớp: L01
Khoa: Khoa học và Kỹ thuật Máy tính

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0.1 Introduction

0.1.1 Problem Statement

In modern daily life, particularly for students, office workers, and couples, the question “What should we eat today?” has evolved from a simple inquiry into a daily source of stress. This issue manifests in three major pain points:

- **Decision Fatigue:** On average, individuals spend 15 to 20 minutes aimlessly scrolling through food delivery applications without making a final choice, often resulting in choosing a random, unsatisfactory meal or skipping it entirely.
- **Choice Overload (The Paradox of Choice):** Existing platforms display hundreds of restaurant options simultaneously in a list format. This overwhelming amount of information causes cognitive overload, making it harder for users to decide and leaving them less satisfied with their eventual choice.
- **Social Friction:** Deciding on a dining location is a common source of conflict for groups, frequently leading to frustration or uneven compromises.

0.1.2 Objectives and Purpose

The primary objective of this project is to develop **Tinner** (a portmanteau of “Tinder” and “Dinner”), a mobile application designed to solve the “Analysis Paralysis” problem in food selection. Guided by the slogan “*Swipe To Bite*”, Tinner simplifies the decision-making process.

The core purposes of the application include:

- **Streamlining Decisions:** Helping users transition from opening the application to deciding on a meal in under 5 minutes.
- **Visual-First Experience:** Providing a “Single-View Interface” that displays one high-quality dish image at a time, removing distractions and stimulating the appetite.
- **Gamified Discovery:** Implementing a swipe-based mechanism (Swipe Left to pass, Swipe Right to save/like) that transforms a tedious chore into an engaging activity.
- **Seamless Information Access:** Upon liking a dish, instantly presenting a Restaurant Info Card with crucial details (address, opening hours, map integration, and reviews) to finalize the decision.
- **Actionable Management:** Allowing users to maintain a personal “Want to Try” list to review, compare, and navigate to selected restaurants efficiently.

0.1.3 Real-world Need and Potential Impact

Comprehensive market research indicates a significant gap in the current ecosystem of food-related applications. While delivery super-apps excel in logistics, social media platforms excel in visual inspiration, and mapping services excel in location data, there is no dominant hybrid model optimized purely for fast, enjoyable, and structured restaurant discovery.

Tinner addresses this real-world need by positioning itself as a “Decision Engine” rather than a transactional middleman. By focusing on the discovery phase and incorporating features like Smart Filters and Personal Collections, the application reduces cognitive load and bridges the gap between structured information and engaging user experience.

The potential impact of Tinner is substantial, especially among Gen Z and Millennials. It promises to drastically reduce the time wasted on daily dining decisions, alleviate the social friction associated with group choices, and provide an intuitive and fun solution to an everyday dilemma.

0.2 Approach and Methodology

0.2.1 Development Process

To ensure the rapid delivery of a Minimum Viable Product (MVP) and allow for iterative improvements, the project adopted an Agile development methodology. This approach enabled the team to prioritize core features—specifically the discovery and decision-making functionalities—while maintaining the flexibility to incorporate user feedback. The development lifecycle was divided into focused sprints, allowing for continuous testing, refinement of the user interface, and validation of the “Visual-First” design philosophy.

0.2.2 Collaboration Tools

Effective communication and version control were maintained using industry-standard collaboration tools. Git and GitHub were utilized for source code management, enabling the team to handle concurrent feature development, conduct code reviews, and track issues seamlessly. Documentation, such as user flows, market research, and feature matrices, were maintained in Markdown formats within the repository to ensure all team members had access to a single source of truth.

0.2.3 Decision-making Rationale

The feature selection and design architecture were driven by the goal of solving specific user pain points, as outlined in the Product Rationale:

- **Focusing on Core Values (Discover & Decide):** The priority was not to build a static directory, but a dynamic tool that directly ends user indecision. Features were evaluated based on an Impact vs. Effort Matrix, prioritizing high-impact functionalities like the Info Card and Personal Collections.
- **“Visual-First” Philosophy:** Operating on the principle that “people eat with their eyes,” the design heavily prioritizes vibrant food imagery over text-heavy menus. This builds trust and evokes emotion at first glance.
- **Solving the Paradox of Choice:** By displaying only one dish at a time and employing a gamified swipe mechanism, the application intentionally limits simultaneous options to reduce cognitive load and turn decision-making into an entertaining experience.

- **Retention Strategy:** The “Want to Try” list (Personal Collections) was implemented as the primary hook. When users invest effort into saving their favorite spots, they create a valuable personal database, significantly increasing the application’s Retention Rate.

0.3 Application Features and Functionality

0.3.1 Core Features (MVP)

The Minimum Viable Product focuses on delivering the essential user flow: discovering dishes, saving preferences, reviewing choices, and navigating to the destination.

- **Swipe to Discover (Swipe-to-Match Logic):** This is the primary discovery mechanism. The system fetches nearby dishes based on the user’s location and presents them in a Single-View Interface. Users swipe left to skip or swipe right to save the dish. This mechanism enforces fast, intuitive decision-making.
- **Manage “Want to Try” List (Personal Collections):** A dedicated space where users can review and manage dishes they previously saved via swiping. It allows users to compare options, reconsider choices, and remove items, transforming the discovery phase into an actionable decision-making process.
- **Restaurant Info Card:** Upon interacting with a dish, the app displays a comprehensive card featuring high-quality food images, menu summaries, opening hours, and highlighted reviews. This provides a quick overview without requiring multiple screen transitions.

0.3.2 Advanced Functionalities

To elevate the user experience beyond basic discovery, several advanced functionalities are integrated into the platform:

- **Smart Filters:** Users can narrow down search results precisely based on real-time needs by filtering by radius (distance), price level, cuisine type, and ratings.
- **Map & Navigation Integration:** Users can view liked restaurants on a map or navigate directly to the location with one tap via Google Maps or Apple Maps integrations, optimizing travel routes.
- **3rd-Party API Integration:** To increase data credibility and provide objective perspectives, the application integrates with external sources (e.g., Google Places, Yelp) to display verified scores and reviews.
- **Group Match and Gamification:** Features like “Group Match” (where a match happens only when everyone in a group swipes right) and “Lucky Spin” are designed to resolve group dining conflicts and turn choosing a restaurant into a fun social game.
- **Food Tours:** The application supports curated itineraries that turn a standard dinner into a micro-adventure, stimulating curiosity and increasing the time users spend engaged with the platform.

0.4 Implementation Details

0.4.1 Tools, Frameworks, and Languages

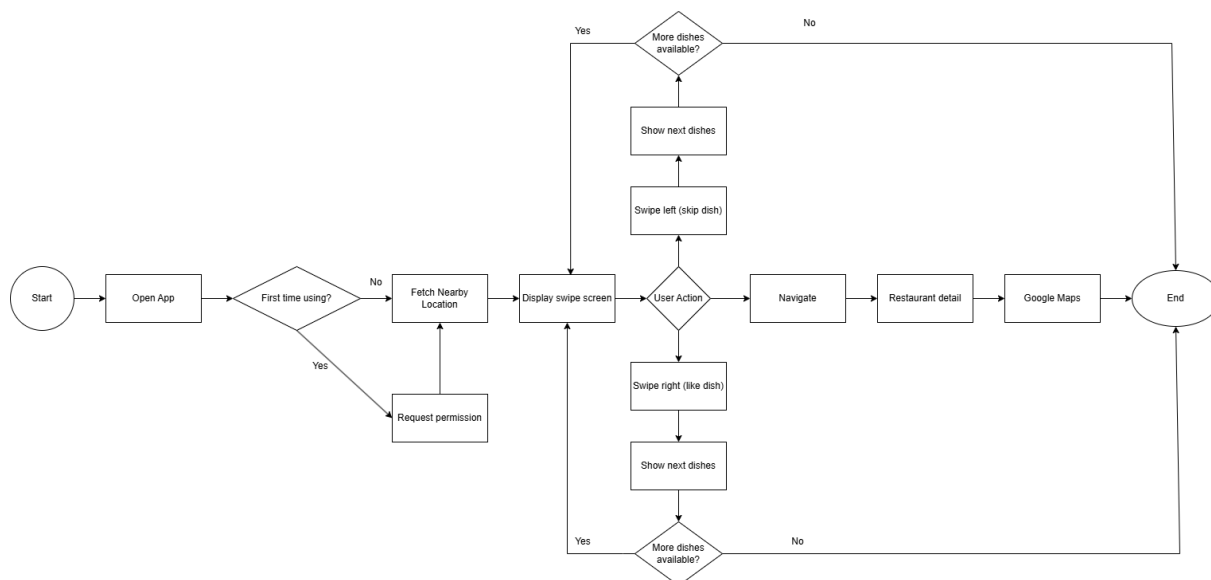
0.5 User Experience

0.5.1 Detailed User Flow

The user flow for the Tinner Minimum Viable Product (MVP) is designed to be intuitive and engaging, focusing on two primary journeys: discovering dishes and managing saved choices.

0.5.1.1 Feature 1: Swipe to Discover and Save Dishes

This feature serves as the primary discovery mechanism, presenting users with nearby food options one at a time to reduce cognitive load. The flow is as follows:



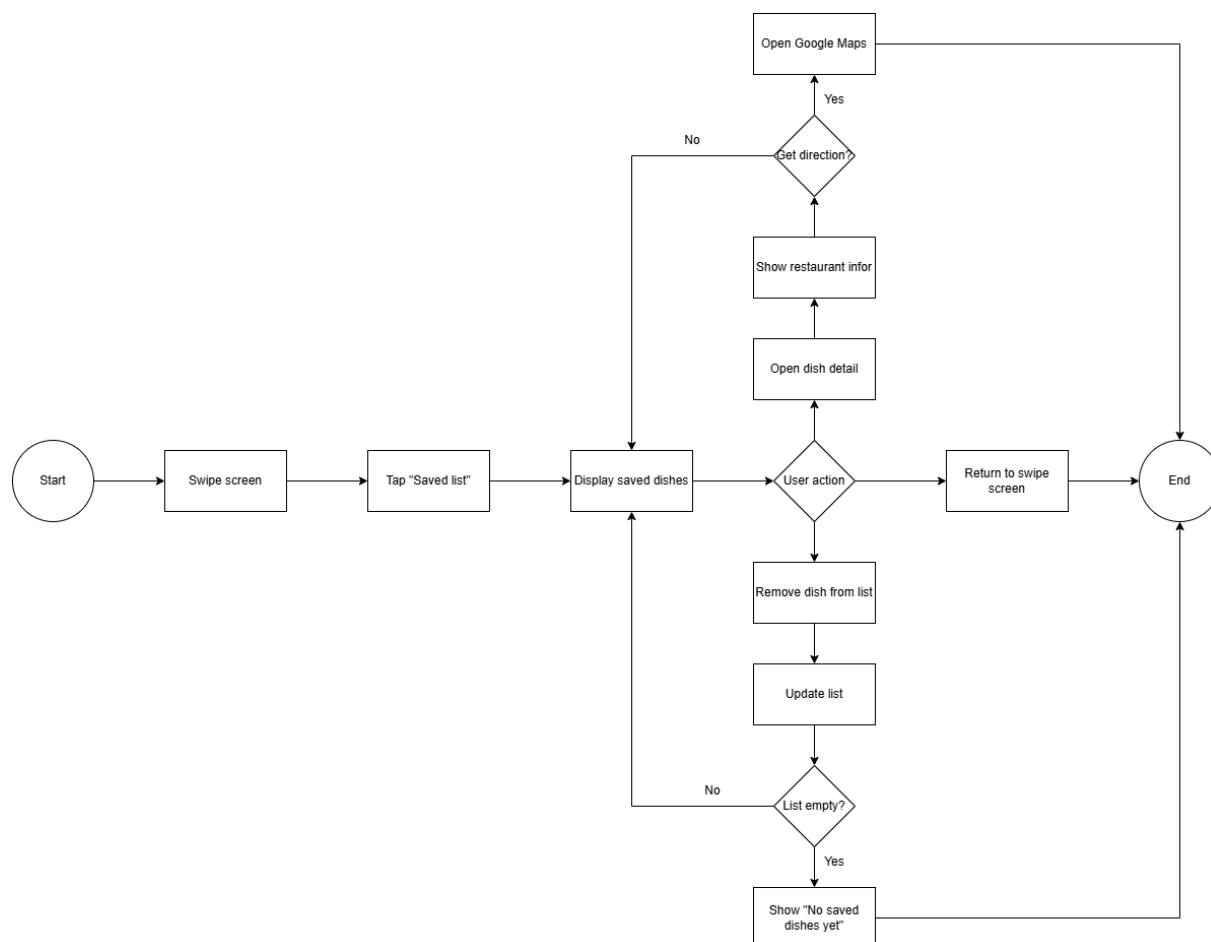
Hình 1: Feature 1: Swipe to Discover and Save Dishes

- Initialization:** The user opens the application. If it is their first session, the system requests location permissions to fetch nearby dining options.
- Discovery Screen:** The system displays a dish card featuring a high-quality image, dish name, price, rating, distance, and the restaurant's name.
- User Actions:** The user evaluates the dish and can perform one of three interactions:
 - Swipe Left:** Skip the dish.
 - Swipe Right:** Save the dish to their personal "Want to Try" list.
 - Tap Navigate:** View detailed restaurant information and open Google Maps for immediate directions.

4. **Continuity:** After an interaction, the system instantly presents the next available dish. If no more dishes match the current location or filters, an “end of stack” notification is displayed.

0.5.1.2 Feature 2: Manage “Want to Try” List

This feature transforms the initial discovery phase into an actionable decision-making process by allowing users to review their liked items.



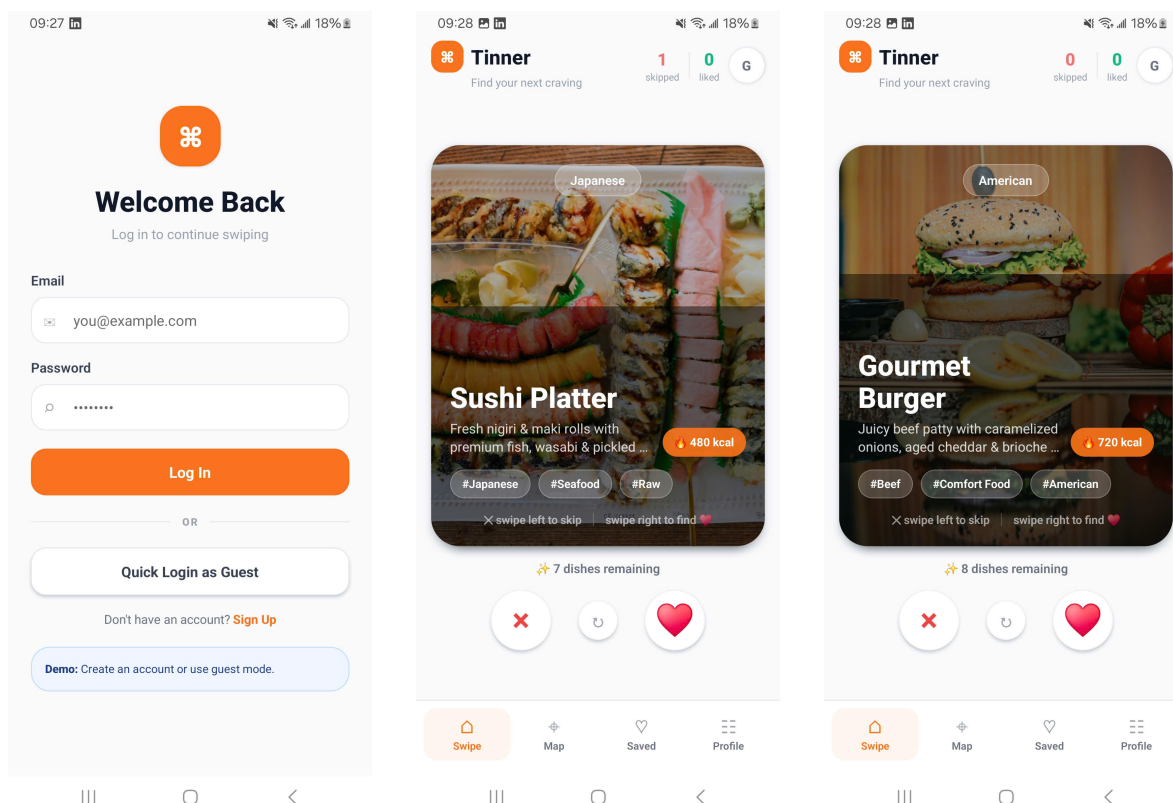
Hình 2: Feature 2: Manage “Want to Try” List

1. **Accessing the List:** The user navigates to the “Want to Try” section from the main swipe interface.
2. **Reviewing Options:** The system presents a structured list of all previously saved dishes, enabling direct comparison.
3. **Decision Actions:** The user can manage their selections by:
 - **Viewing Details:** Accessing the comprehensive dish and restaurant detail screen to get directions via mapping services.
 - **Removing Items:** Deleting a dish from the list if they lose interest. The system handles empty states gracefully if all items are removed.

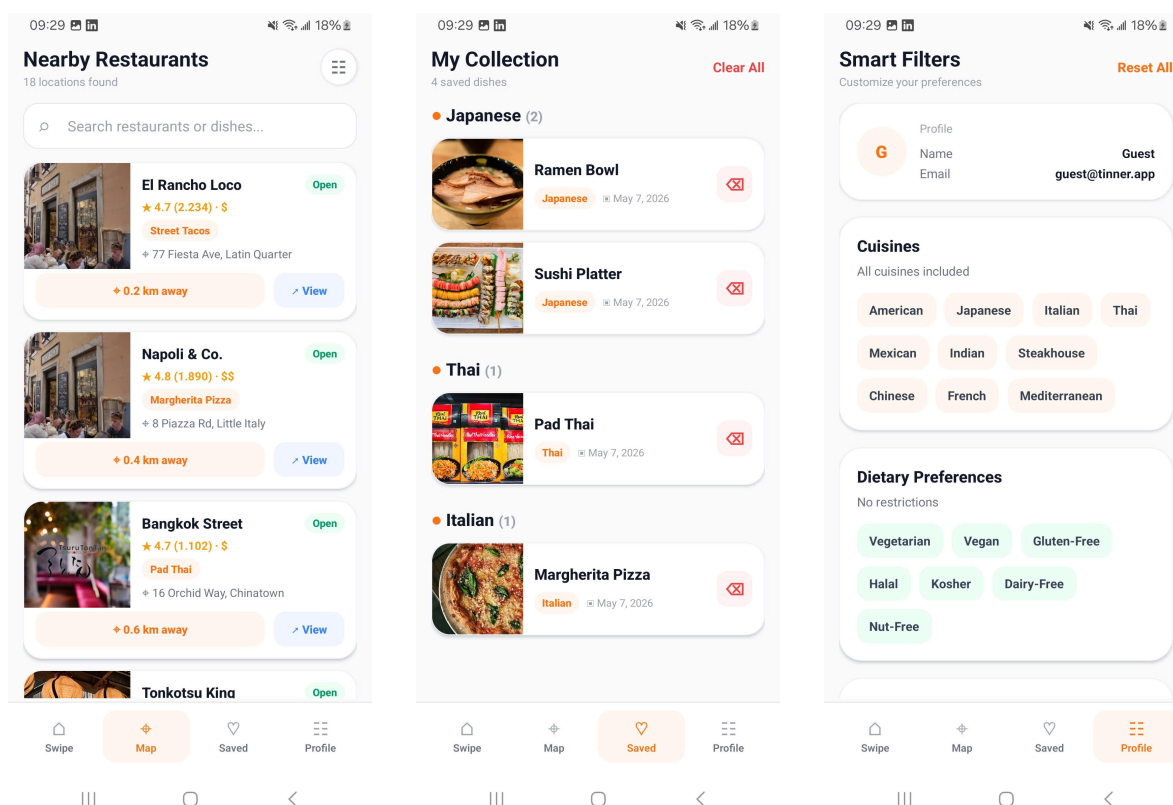
- **Returning:** Navigating back to the main swipe screen to continue discovering new food.

0.5.2 Screenshots

The following screenshots illustrate the core navigation and user interactions within the application, demonstrating the "Visual-First" philosophy and intuitive user interface.



Hình 3: From left to right: Login Screen, Discovery (Swipe) Interface displaying dish information, and Swipe Interaction.



Hình 4: From left to right: Integrated Map & Navigation, “Want to Try” (Saved) List, and User Profile.

0.6 System Architecture

0.6.1 Architecture Overview

0.7 Web Service Design

0.7.1 Backend Services

0.7.2 Best Practices

0.8 Deployment Strategy

0.8.1 Application and Backend Deployment

0.8.2 CI/CD Pipelines

0.9 Testing Coverage Report

0.9.1 Code Coverage Metrics

0.9.2 Gap Analysis and Code Quality

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0.11.1 Successes and Challenges

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0.13 Product Distribution and Survey

0.13.1 Promotion and Marketing

0.13.2 Survey Design and Analysis

0.13.3 App Metrics Collector

0.13.4 Feedback Evaluation and Behavior Analytics